IDE 402

DESIGN MILESTONES

WHISKA

derived from the word “whisk,” the essential tool used to prepare matcha, with the reference of matcha.

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Milestone 1 - Research

Early Concepts

- **Why do all options require special preparation stages?**
  - *Interviewees: "I enjoy eating and drinking things much more when they involve a ritual and allow me to participate in the preparation process."*
- **What are the common motivations of potential users?**
  - *Interviewees: "Emotional comfort*
    Health and wellness optimization
    Small daily ritual in a fast-paced life
    Stress relief during urban life."

	Rejected Options		Selected Option
	Concept 1: Hot Chocolate Machine	**Concept 2: Herbal Tea Machine**	**Concept 3: Matcha Machine**
Why?	*No market demand.*	*Every herbal tea request special treatment ı wanted to focus on one specific beverage type.*	*Identified a gap in the market for this specific beverage.*

Research

Observation: The ways in which businesses prepared and served matcha were observed.

Method	Texture	Foam	Taste	Speed
Chasen	Creamy	Fine micro-foam	Balanced	Medium
Milk Frother	Lighter	Large bubbles	More diluted	Fast
Blender	Dense-uneven	Thick foam	Sharper	Very fast

Observation Insight: *Matcha is being prepared using the wrong methods. Incorrect and quick methods are ruining its taste and texture.*

Existing Products Benchmark:

Top 2 Tensions:

- *Incorrect matcha production in the market.*
- *People are willing to pay for a cup of valuable matcha and than be disappointed.*

Opportunity Areas:

- *Even when culinary matcha is used, the goal is to achieve the best possible quality.*
- *A system can be designed to sift, whisk and heat the water to the ideal temperature automatically..*

Milestone 2 - Research/Concept Selection

Selected Main Concept: A commercial-grade matcha machine that preserves the ceremonial essence of traditional preparation while optimizing speed, consistency, and workflow efficiency for modern hospitality environments.

User Persona

- Location: Urban city
- Lifestyle: Urban professional, health-conscious, busy daily routine

Goals

- Drink high-quality matcha every day
- Experience the ritual and mindfulness of preparing matcha
- Save time without compromising quality

Frustrations

- Most cafés prepare matcha incorrectly or with
- poor technique
- Matcha drinks outside are expensive and inconsistent
- Preparing matcha traditionally at home requires time and specific tools
- Many machines simplify the process but remove the ritual aspect

Needs

- A consistent and high-quality matcha preparation method
- A product that preserves the **traditional whisking** process
- A solution that fits into a busy urban lifestyle
- A machine that combines ritual, precision, and convenience

Properties of my matcha:

- Sifted, smooth, clump-free matcha powder
- Gradual addition of water at the correct temperature
- An automatic mixer that follows a W shape

with the help of a whisk made of bamboo or alternative material.

What ı want to achieve?

- To serve fast
- To serve quality
- To serve with awareness-respect of matcha ritual
- To show the ritual to people with transparency

Research

History of Matcha

Period	Country	How Was It Consumed?
Tang-Song Dynasty	China	Powdered tea, whisked
12th Century	Japan	Zen meditation drink
16th Century	Japan	Aesthetic tea ceremony

How to make the perfect matcha?

Initial Sketches

Milestone 3 - Selected Design Prototype

Milestone 3 - Selected Design Prototype

Key Machine Components and Materials

1. Matcha powder chamber
2. Water Reservoir System
3. Matcha Dosing Channel
4. Vibrating Sifter Unit
5. Heated Water System
6. Chawan Holder & Lift Mechanism
7. Mechanical Whisk Unit
8. Integrated Water Delivery System
9. Ceramic Chawan Bowl
10. Control Interface

Usage Scenario

Google Gemini - AI Prompt

The machine has a symmetrical vertical form with two side columns and a central preparation area. The design is minimal, calm, and inspired by the aesthetics of traditional tea ceremonies.

Structure of the Machine

At the top center of the machine there is a lidded compartment where matcha powder is added.

Behind this compartment there is another hidden lid for filling the water reservoir.

The right and left vertical columns of the machine contain water tanks that supply hot water during preparation.

At the bottom front of the machine there is a control button that allows the user to select how many portions of matcha to prepare.

In the center of the machine there is a ceramic matcha bowl (chawan) placed inside a vertical lift mechanism.

Above the bowl there is a mechanical whisk unit inspired by the traditional bamboo chasen.

Preparation Process

The user fills the matcha powder compartment at the top.

The user fills the water reservoir located behind the powder compartment.

The user places a ceramic chawan bowl into the preparation slot.

The user presses the portion selection button.

The machine begins the process.

First, a precise dose of matcha powder is released through a small channel under the whisk housing and falls into the chawan bowl.

Then the bowl rises upward using a vertical elevator mechanism.

At the same time, the whisk unit moves forward approximately 10–15 cm so it can position itself inside the bowl.

Once the whisk is inside the bowl, hot water is released through the center of the whisk.

The whisk then begins a linear "W-shaped whisking motion", mimicking the traditional matcha whisking technique used in tea ceremonies.

This motion dissolves the matcha powder and creates the characteristic smooth and foamy texture.

When the whisking process is finished, the bowl descends back down using the same elevator mechanism.

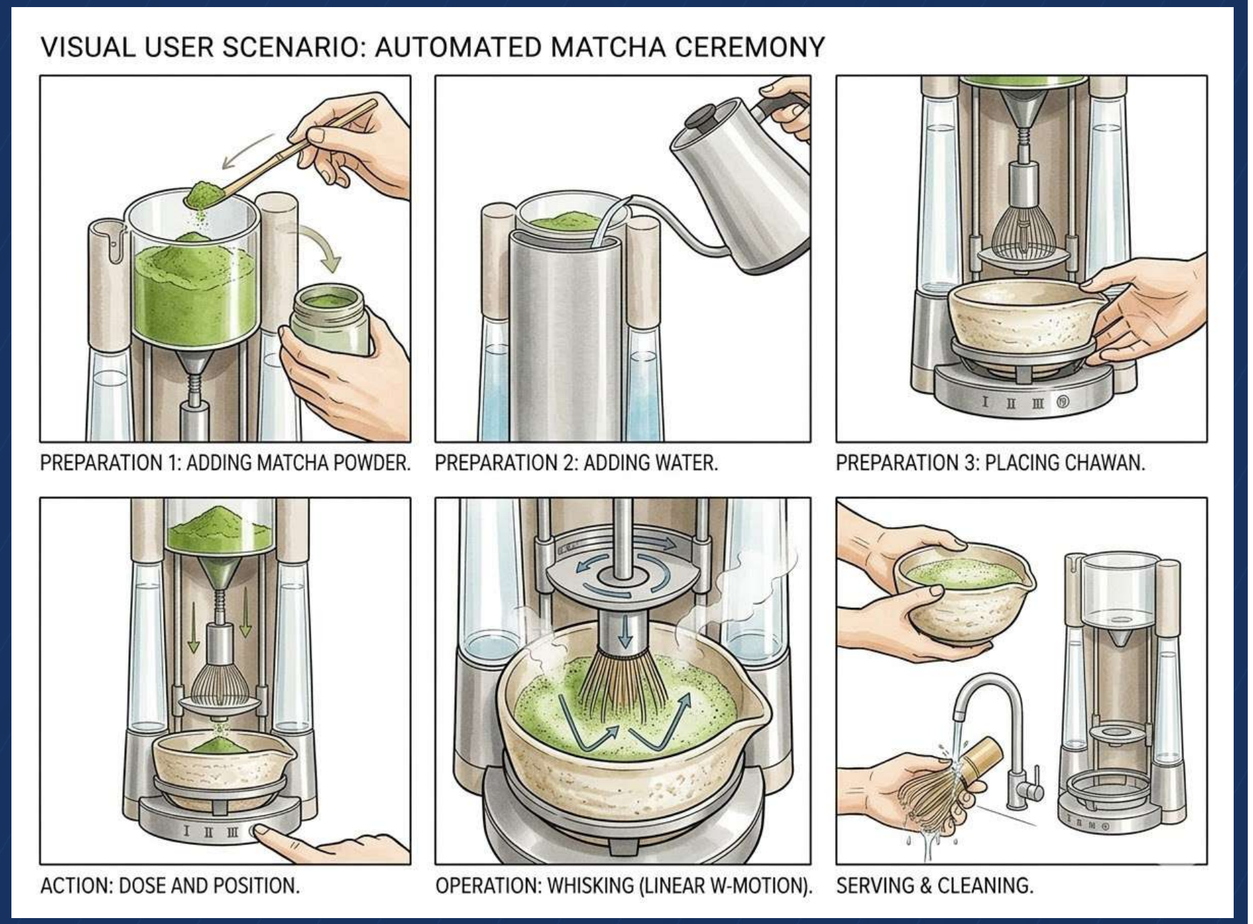
Serving

The user manually removes the chawan bowl and pours the prepared matcha into serving cups.

Cleaning

The whisk unit can be manually detached, washed, and reattached.

After cleaning, the machine is ready for the next preparation cycle.



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Thank you...

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